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How to Read a Vernier (caliper): 5 Steps

5-7 minutes

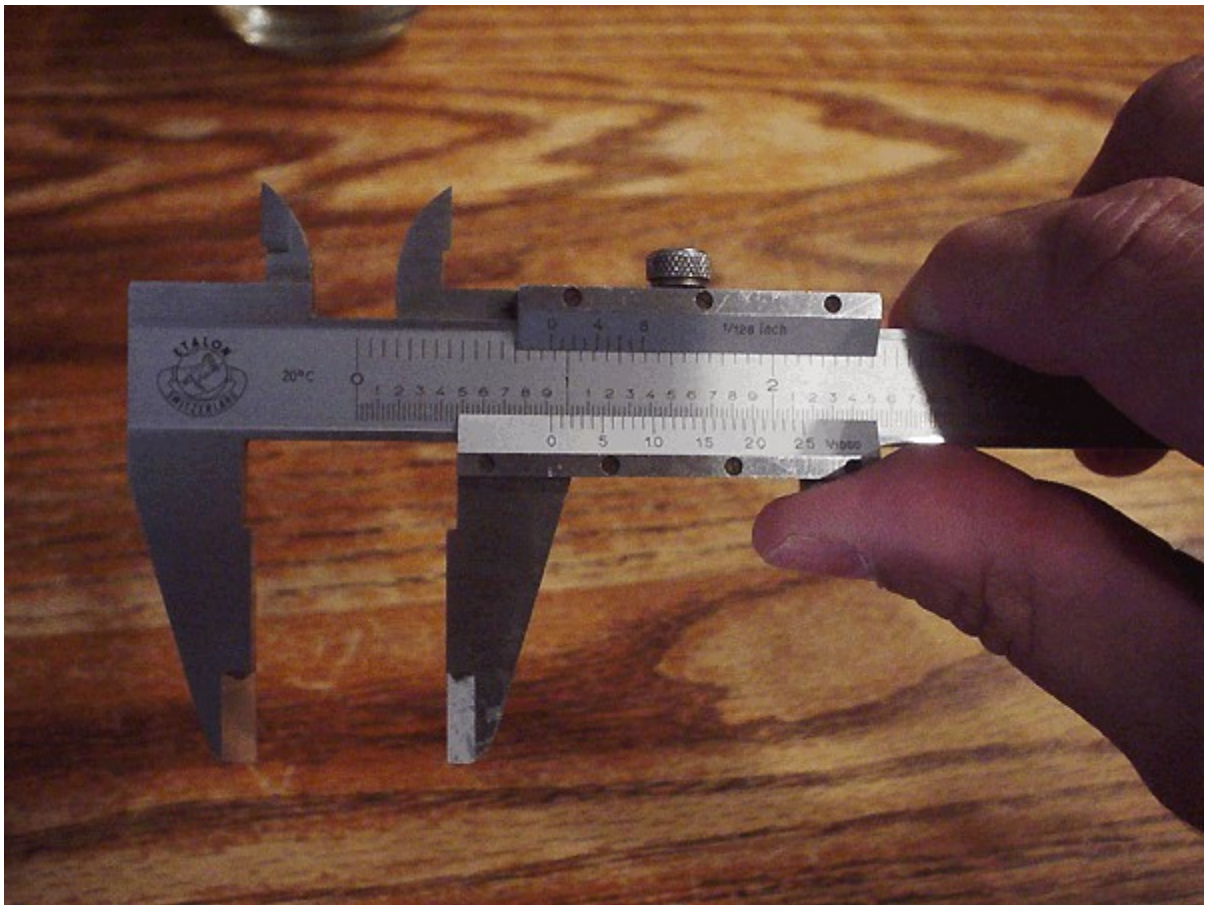
Introduction: How to Read a Vernier (caliper)



Looking at an early caliper, you'll notice that it kinda looks like a tomahawk and has a scale down it's length. More about the scale later. On the bottom left, the "blade of the hatchet" are the jaws of the caliper for measuring **outside dimensions** - think of a tube. This would measure the outside diameter. The "pike" on top of the hatchet is used for measuring **inside dimensions** . Again, think of a tube's inside diameter. Looking all the way to the right...I know it's hard to see...but there's a little "tail" sticking out of the center of the

handle. That will measure the **depth** of a step or some other similar feature. That little button on the top near the "pike" is a locking screw. The one just below it is a thumb button.

Step 1: Holding the Caliper

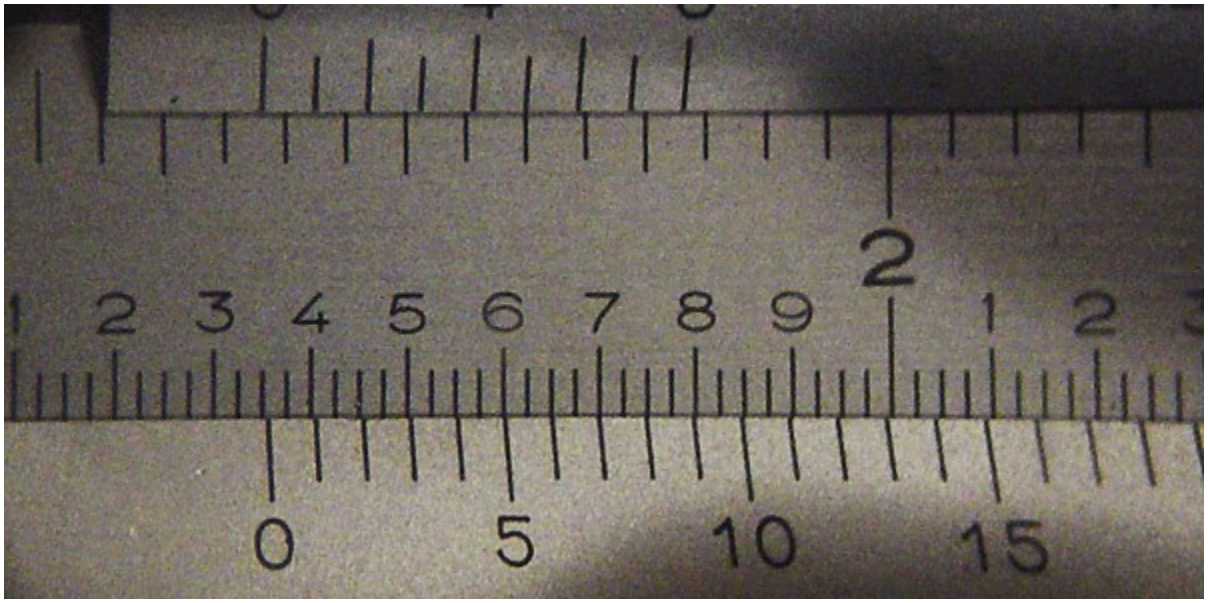


The inside measurement here...

is EXACTLY the same as a measurement here.

This is the proper way to hold a caliper. Use your thumb to slide the jaws open and closed. Yup, another tool that was totally designed for righties. Use the locking screw at the top if you want to set a dimension on the caliper and then check parts against your setting. Though it may not look it, the measurement between the inside and outside jaws are EXACTLY the same.

Step 2: Course Reading #1



these numbers divide the inch into 10 equal parts

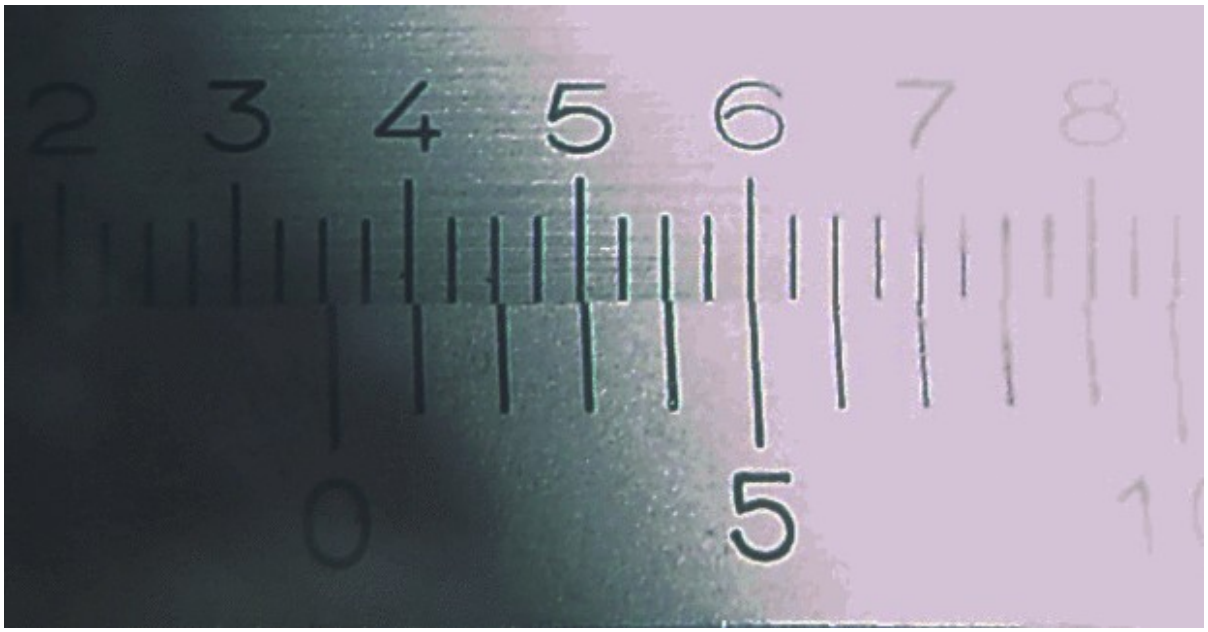
So let's read a measurement off of the caliper.

First, a quick math refresher. Whole numbers (1,2,3, etc.) are followed by a decimal point and an endless number of zeros (2.000000). We are going to be working with the 3 places to the right of the decimal point. The first number to the right of the decimal point is a tenth of an inch (1/10), the second number signifies a hundreth of an inch (1/100), and the third number signifies a thousanth of an inch (1/1000). A human hair is about 3 thousands of an inch (0.003), give or take a thousanth. This number (0.456) is read as four hundred and fifty-six thousands.

If you noticed, the "handle" of the caliper has a scale that starts from zero on the left and goes up to 6 inches on the right, in this case. Other calipers can be much longer in length. The big number in the center of the scale is the inch number. In this case we are looking mostly at the area AFTER the first inch mark on the left, and then we see the number **2** on the right, which indicates the

second full inch. Notice that each inch is divided into 10 equal parts with the numbers 1 through 9. That makes each one of those lengths 1/10 of an inch (0.100). And then each amount of space between each 1/10 of an inch has 3 lines - which breaks each 1/10 of an inch into 4 equal pieces. Each one of those little pieces, therefore, is twenty-five thousanths of an inch (0.025) long. There's also a scale on the bottom of the picture. It starts at zero, there's a 5, a 10 and a 15. It goes up to 25, but the image doesn't show that. THAT, is the vernier part of the caliper. Notice that the line for the 0 (zero) is to the right of the second line after the number 3. Since we know that there is a large number 1 to the left of the image that we can't see (which signifies 1 whole inch), let's write that down (1.) and we'll put the decimal point in. The 3 is to the left of the zero on the vernier, so we'll add that to our number (1.3). The zero line is to the right, of the second line after the 3. So there is at least fifty thousanths there (0.050), but there's more. Notice how the zero line is just a smidge to the right of that second line after the 3. Now look at the large 2 and notice that the vernier line of 13 is to the LEFT. So the trick is to find which lines on the vernier lines up exactly with the scale on the center. In this case, it looks like 6...but let's get a closer look.

Step 3: Fine Reading #1

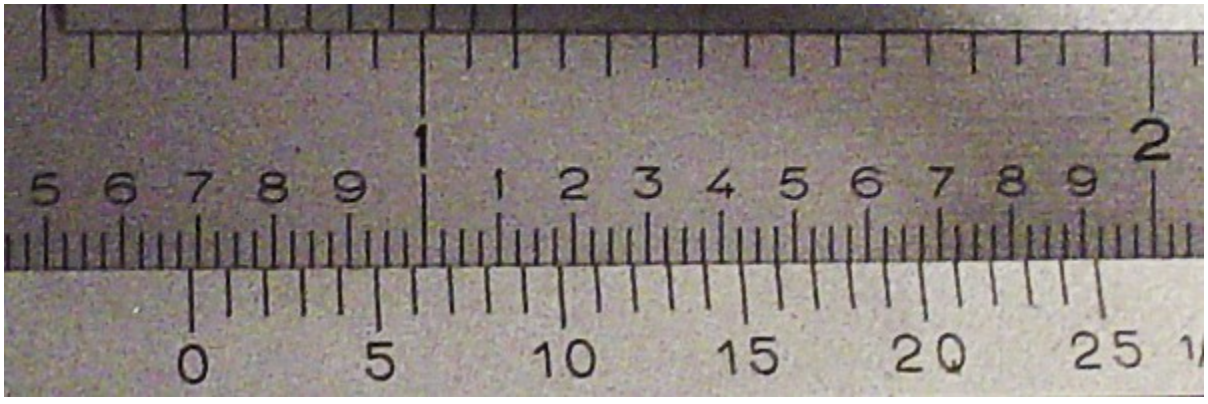


Yup. It's 6 all right.

NOT the 6 on the top scale - the 6 on the bottom vernier scale.

So, we were up to 1.3, plus we knew we had at least another fifty thousanths (0.050), and the vernier tells us there is an addition six thousanths (0.006). Add them all up and we have 1.356 - read 1 inch, three hundred and fifty-six thousands.

Step 4: Course Reading #2



Let's try another one.

The zero on the vernier is to the left of the 1 whole inch mark...so the number is LESS THAN 1 full inch. Let's write that down (0.) and don't forget the decimal point.

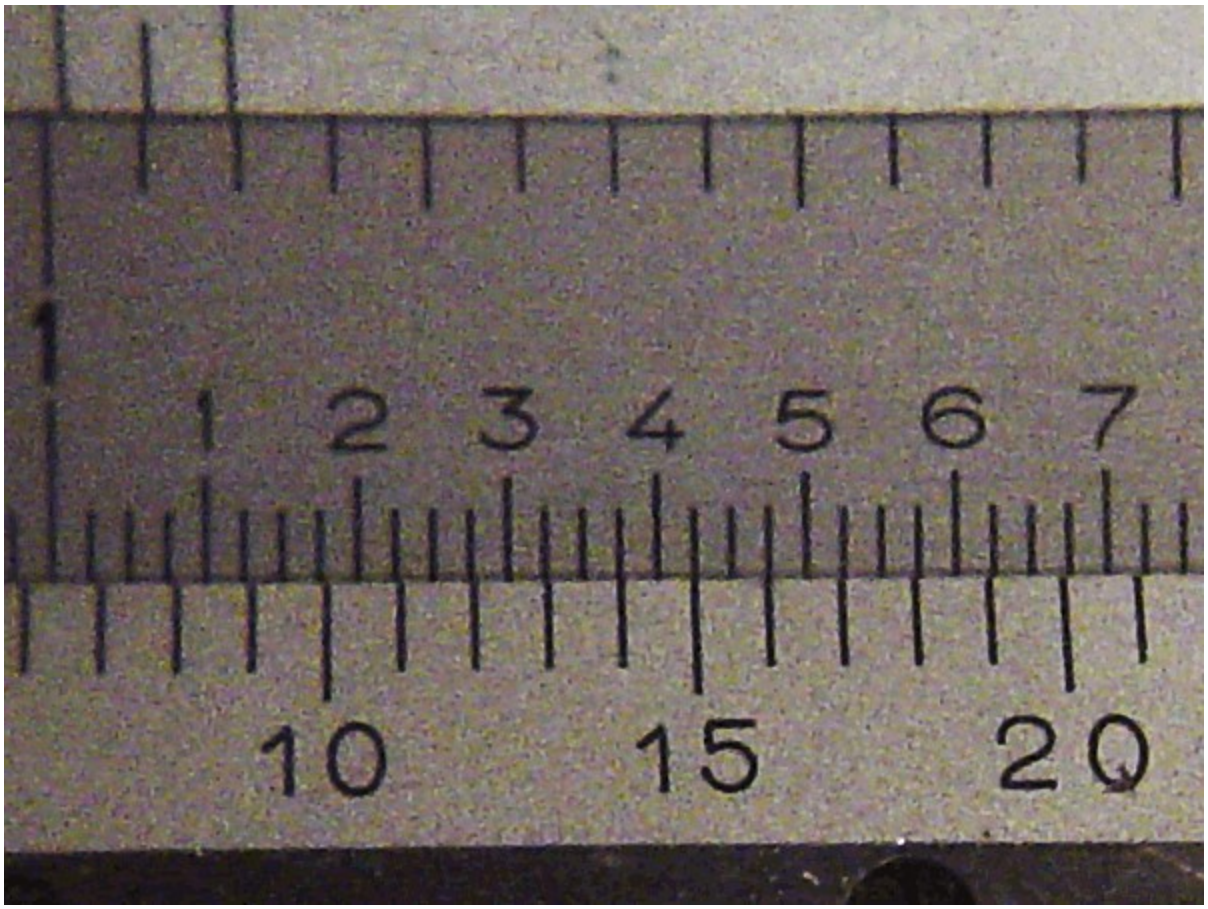
We're between 6 and 7 on the 1/10 scale, so let's write that down too (0.6).

Hmmm, the vernier zero line is almost at the number 7; there are 3 lines...seventy-five thousanths (0.075)...to the left of the vernier zero line...and some more.

Let's look at the vernier scale and see what lines up to understand how much more.

16, 17, 18 - not sure. Let's get a better shot of that.

Step 5: Fine Reading #2



It looks like 15.

So, that's 0.6 plus 0.075 plus 0.015.

That would be 0.690.

Read, six hundred and ninty thousanths.